

```

    value: eks-cluster
  - name: PATH_TO_K8S_YAML
    value: /workspace/source/
path/to/kubernetes-deployment.yaml

```

The necessary Tekton tasks (git-clone, maven, buildah, docker-push, and eks-deploy) are defined in your cluster (pulled from tektonhub), after you have set up the appropriate credentials for accessing the Git repository and Docker registry.

Additionally, the Kubernetes deployment YAML should be available in the Git repository and specified correctly in the PATH_TO_K8S_YAML parameter.

```

apiVersion: tekton.dev/v1beta1
kind: Pipeline
metadata:
  name: java-maven-docker-pipeline
apiVersion: Specifies the Tekton API
version being used.
kind: Indicates that this resource is a
Tekton Pipeline.
metadata.name: The name of the pipeline
resource.
spec:
  resources:
    - name: source-repo
      type: git
    - name: docker-image
      type: image

```

spec.resources: Defines the resources used by the pipeline.

resources.name: Names the resources. In this case, there's a Git resource named *source-repo* and an image resource named *docker-image*.

resources.type: Specifies the type of resource, which is either git for the Git resource or image for the Docker image resource.

These sections define the individual tasks that make up the pipeline:

```

tasks:
  - name: clone-repo
    taskRef:
      name: git-clone

```

```

resources:
  inputs:
    - name: url
      resource: source-repo
    - name: revision
      resource: source-repo
  params:
    - name: path
      value: /workspace/source

```

tasks.name: Name of the task.

tasks.taskRef.name: Refers to the name of the Tekton task that will be executed (in this case, git-clone).

tasks.resources.inputs: Specifies the inputs required for this task. The URL and revision are inputs defined in the Git resource, and the path is set to the working directory for the task.

```

  - name: build-with-maven
    taskRef:
      name: maven
    resources:
      inputs:
        - name: workspace
          resource: source-repo
      params:
        - name: PATH_TO_POM
          value: /workspace/source/path/
to/pom.xml
        - name: GOALS
          value: "clean package"

```

This task (build-with-maven) runs the Maven Tekton task. It uses the Git resource *source-repo* as its workspace.

The Maven build is configured with the path to the *pom.xml* file and the goals 'clean package'.

The remaining tasks (build-

docker-image, push-to-docker-registry, and deploy-to-eks) follow a similar structure. They reference specific Tekton tasks and define the resources and parameters required for each step.

```

Pipelinerun.yaml
apiVersion: tekton.dev/v1beta1
kind: PipelineRun
metadata:
  name: java-maven-docker-pipeline-run
spec:
  pipelineRef:
    name: java-maven-docker-pipeline
  resources:
    - name: source-repo
      resourceRef:
        name: your-git-resource-name
    - name: docker-image
      resourceRef:
        name: your-docker-image-
resource-name

```

To run the above pipeline:

Create pipeline: Apply the pipeline definition YAML to your cluster using *kubectl apply -f pipeline-definition.yaml*.

Create pipeline run: Apply the pipeline run YAML to start the execution of your pipeline using *kubectl apply -f pipeline-run.yaml*. Make sure you adjust the resource names (Git and Docker image resources) in the pipeline run YAML.

Monitor progress: Use commands like *kubectl get pipelineruns* and *kubectl describe pipelinerun <run-name>* to monitor the progress of your pipeline run. You can also use *kubectl logs -f taskrun <task-run-name>* to view the logs of individual tasks. 

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By: Venkateswaran N.

The author is a lead architect, containerisation and modernisation, AWS practice, IBM India.